

AORC Data Retrieval Notes

Homa Salehabadi, David Tarboton

1 Introduction

The Analysis of Record for Calibration (AORC) dataset is a high-value dataset of interest to the CUAHSI and CIROH community. It is hosted in Amazon web services in multiple formats. We have developed Jupyter Notebooks to enable access to this data from a JupyterHub computing platform to improve the accessibility of this data.

2 Datasets

There are two primary sources for the AORC dataset: 1- latitude-longitude gridded dataset, and 2- the dataset for the National Water Model (NWM) in the projected coordinate system of the NWM.

2.1 Latitude-Longitude Gridded AORC Dataset

The latitude-longitude version 1.1 of the AORC dataset is hosted on Amazon S3 bucket (s3://noaa-nws-aorc-v1-1-1km) and is publicly accessible via the AWS Open Data Registry on <https://registry.opendata.aws/noaa-nws-aorc/>. It is organized first by year and then by variable (see Figure 1 and <https://noaa-nws-aorc-v1-1-1km.s3.amazonaws.com/index.html>).

The latitude-longitude gridded AORC dataset version 1.1 uses a geographic (longitude-latitude) World Geodetic System 1984 (WGS 84) coordinate system, with a spatial resolution of 30 arc seconds. The spatial coverage of the dataset includes the contiguous United States (CONUS), as well as portions of southern Canada and northern Mexico (Figure 2). The dataset contains 8401 longitude values ranging from -130 °E to -60 °E, and 4201 latitude values ranging from 20 °N to 55 °N, with the outermost areas (the transparent white rectangle surrounding the blue area in Figure 2) containing NaN values.

The temporal coverage of the dataset spans from February 1979 (1979-02-01 00:00:00) to the present, with a temporal resolution of 1 hour. There are 46 Zarr files corresponding to the years 1979 to 2024. However, using the current version of the Jupyter notebook developed, data can only be retrieved up to 2022-12-31 23:00:00, as it produces errors for 2023 and later).

Table 1 lists the variables in the latitude-longitude gridded AORC dataset.

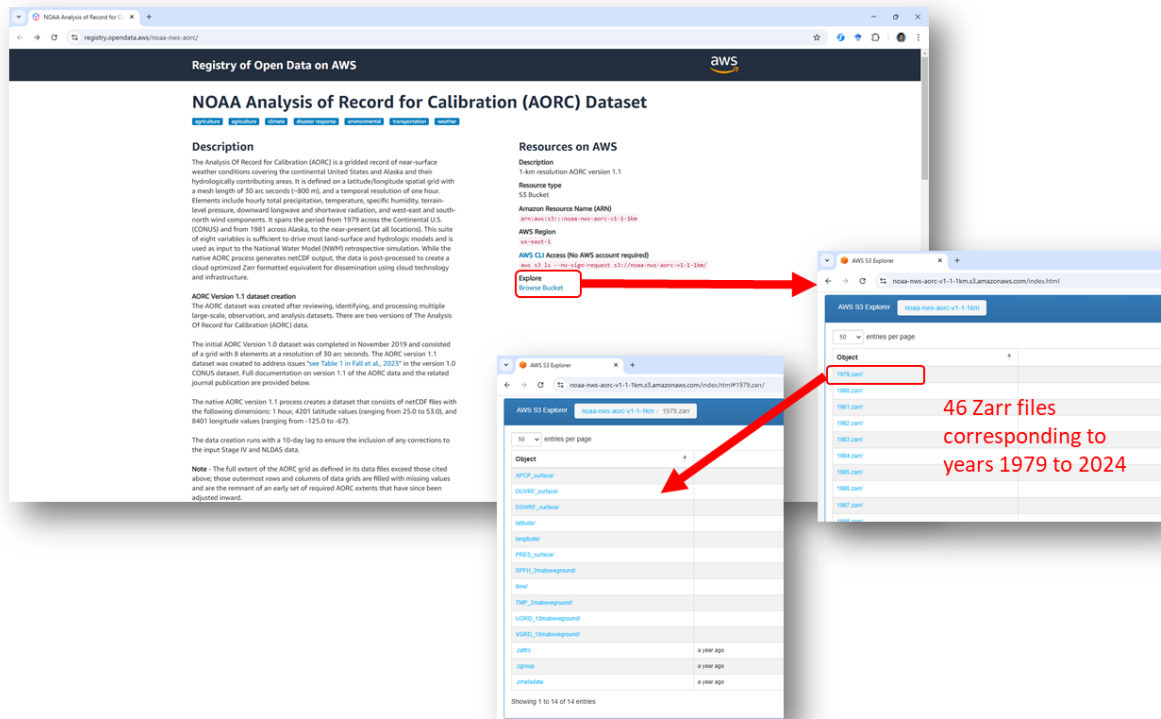


Figure 1. NOAA Analysis of Record for Calibration (AORC) dataset

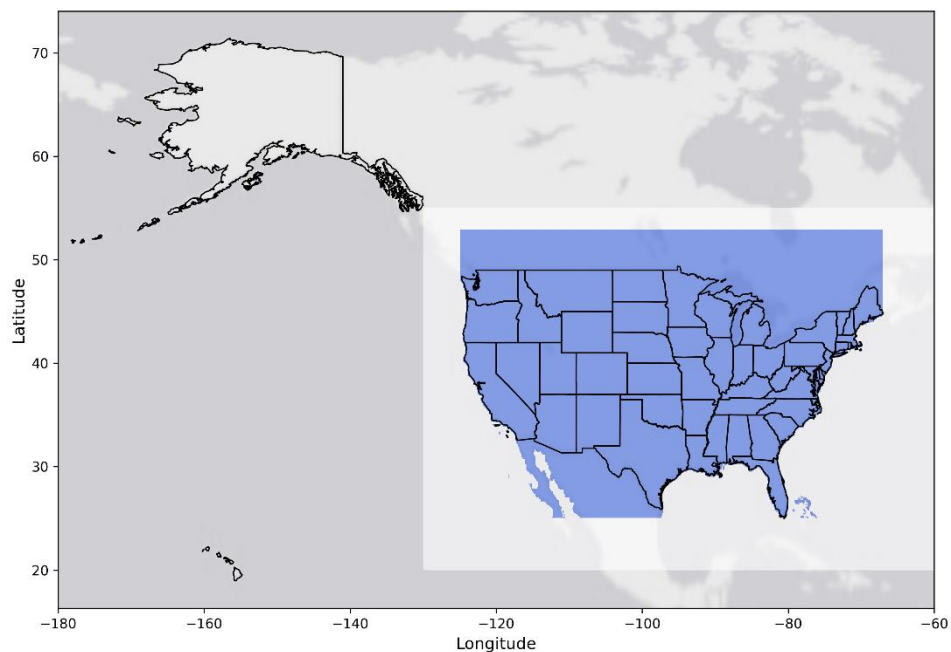


Figure 2. Spatial coverage of the latitude-longitude gridded AORC dataset version 1.1 (blue area). The transparent white rectangle around the colored area represents the NaN values in the dataset.

Table 1. Variables in the Latitude-Longitude gridded AORC dataset

Variable Name*	Long Name	Description
APCP_surface	Total Precipitation	Hourly total precipitation (kgm^{-2} or mm)
TMP_2maboveground	Air Temperature	Temperature (at 2 m above-ground-level (AGL)) (K)
SPFH_2maboveground	Specific Humidity	Specific humidity (at 2 m AGL) (g g^{-1})
DLWRF_surface	Downward Long-Wave Radiation Flux	longwave (infrared) radiation flux (at the surface) (W m^{-2})
DSWRF_surface	Downward Short-Wave Radiation Flux	Downward shortwave (solar) radiation flux (at the surface) (W m^{-2})
PRES_surface	Pressure	Air pressure (at the surface) (Pa)
UGRD_10maboveground	U-Component of Wind	(west to east) - components of the wind (at 10 m AGL) (m s^{-1})
VGRD_10maboveground	V-Component of Wind	(south to north) - components of the wind (at 10 m AGL) (m s^{-1})

* Variable names from this table should be used to specify the variable of interest to retrieve data using the *AORC_LL_PointRetrieval.ipynb* and *AORC_LL_ZoneRetrieval.ipynb* Jupyter notebooks explained in section 3. These are the names used in the Zarr files of the Latitude-Longitude gridded AORC dataset hosted in Amazon Web Services (e.g., `s3://noaa-nws-aorc-v1-1-1km/1990.zarr/APCP_surface` is the directory to total precipitation data)

2.2 NWM-CONUS projected AORC data

This dataset is part of the National Water Model (NWM) Retrospective archive, which includes input and output from multi-decade retrospective simulations, available through AWS Open Data Registry (<https://registry.opendata.aws/nwm-archive/>). There are 8 Zarr files corresponding to 8 variables in <https://noaa-nwm-retrospective-3-0-pds.s3.amazonaws.com/index.html#CONUS/zarr/forcing/> (Figure 3).

The NWM-CONUS AORC dataset uses a projected Lambert Conformal Conic coordinate system. It has a 1 km spatial resolution and contains 4608 X values ranging from -2303499.25 m to 2303500.75 m, and 3840 Y values ranging from -1919500.375 m to 1919499.625 m (Figure 4). There are no NaN values in this dataset. The temporal coverage of the NWM AORC dataset spans from February 1979 (1979-02-01 00:00:00) to January 2023 (2023-01-31 23:00:00), with a temporal resolution of 1 hour.

Table 2 lists the variables in the NWM projected AORC dataset.

Table 2. Variables available in the NWM projected AORC dataset

Variable Name	Long Name	Standard Name	Units
RAINRATE	Surface Precipitation Rate	precipitation_flux	mm s ⁻¹
T2D	2-m Air Temperature	air_temperature	K
Q2D	2-m Specific Humidity	surface_specific_humidity	kg kg ⁻¹
LWDOWN	Surface downward long-wave radiation flux	surface_downward_longwave_flux	W m ⁻²
SWDOWN	Surface downward short-wave radiation flux	surface_downward_shortwave_flux	W m ⁻²
PSFC	Surface Pressure	air_pressure	Pa
U2D	10-m U-component of wind	x_wind	m s ⁻¹
V2D	10-m V-component of wind	y_wind	m s ⁻¹

* Variable names from this table should be used to specify the variable of interest to retrieve data using the *AORC_NWMProj_PointRetrieval.ipynb* and *AORC_NWMProj_ZoneRetrieval.ipynb* Jupyter notebooks explained in section 3. These are the names used in the Zarr files of the NWM projected dataset hosted in Amazon Web Services (e.g., s3://noaa-nwm-retrospective-3-0-pds/ CONUS/zarr/forcing/precip.zarr/RAINRATE is the directory to Surface Precipitation Rate)

3 Jupyter Notebooks for data retrieval and processing

To facilitate efficient data retrieval and processing, we developed four Jupyter Notebooks—two for each dataset—to retrieve data for either a specified point of interest or a defined spatial domain using a shapefile. The notebooks are:

1. **AORC_LL_PointRetrieval.ipynb**: Retrieves and aggregates data from the AORC latitude-longitude gridded dataset for a specific point specified using geographic coordinates.
2. **AORC_LL_ZoneRetrieval.ipynb**: Retrieves and aggregates data from the AORC latitude-longitude gridded dataset for an area defined by a polygon shapefile.
3. **AORC_NWMProj_PointRetrieval.ipynb**: Retrieves and aggregates data from the NWM projected AORC dataset for a specific point within the NWM coverage area, specified using geographic coordinates.
4. **AORC_NWMProj_ZoneRetrieval.ipynb**: Retrieves and aggregates data from the NWM projected AORC dataset for an area within the NWM coverage area, defined by a polygon shapefile.

The AORC datasets can be accessed through these Jupyter Notebooks deployed on computing platforms such as CIROH 2i2c linked to HydroShare. These Jupyter Notebooks provide instructions and code for retrieving the AORC dataset hosted on Amazon Web Services and subsetting and aggregating at a user-defined time scale (which may differ from NOAA’s original time steps) and spatial area.

3.1 Point Data Retrieval Notebooks

AORC_LL_PointRetrieval.ipynb and *AORC_NWMProj_PointRetrieval.ipynb*

The point data retrieval notebooks were created to make it easy for researchers to access AORC data for a specific point within the CONUS area, specified using geographic coordinates. Each notebook takes as inputs the coordinates (e.g. latitude and longitude) of a study location in any coordinate system, start and end dates for the desired study period, a variable name, and a preferred time aggregation interval. The notebooks then retrieve data from Amazon Web Services (s3://noaa-nws-aorc-v1-1-1km for the latitude-longitude gridded dataset and s3://noaa-nwm-retrospective-3-0-pds/ for the NWM projected

dataset), aggregate it over the specified time interval, save the data as a comma separated values (CSV) file, and display it as a plot.

As examples, Figure 6 and Figure 7 illustrate outputs from the *AORC_LL_PointRetrieval.ipynb* notebook, which retrieves data from the latitude-longitude gridded AORC dataset. Figure 6 shows the total precipitation at a point near Salt Lake City, UT (Figure 5), from January 1 to January 31, 1990, aggregated into daily time intervals by summing the hourly data. Figure 7 presents the temperature at the same point from January to December 2021, aggregated to monthly time intervals by averaging the hourly data. The notebook also saves the data as a comma-separated values (CSV) file.

The *AORC_NWMPProj_PointRetrieval.ipynb* notebook produces similar outputs, but the data are retrieved from the NWM projected dataset.

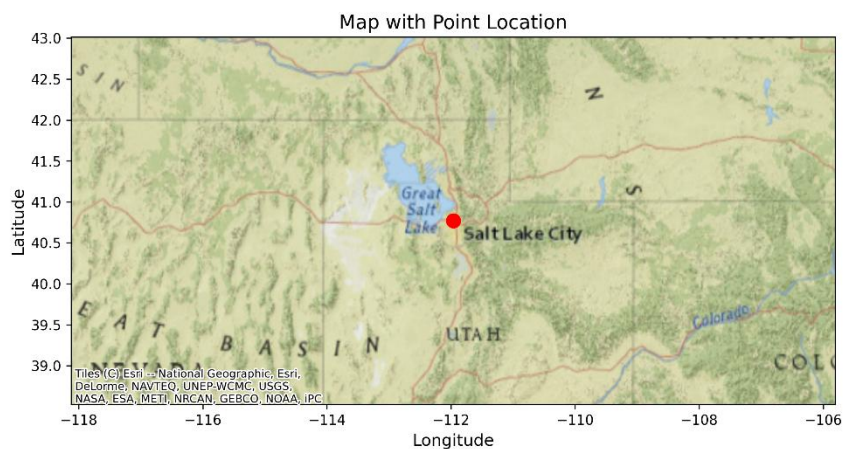


Figure 5. Map showing the example point of interest near Salt Lake City, UT (longitude: -111.96503, latitude: 40.77069, coordinate system: EPSG:4326) used to demonstrate data retrieval.

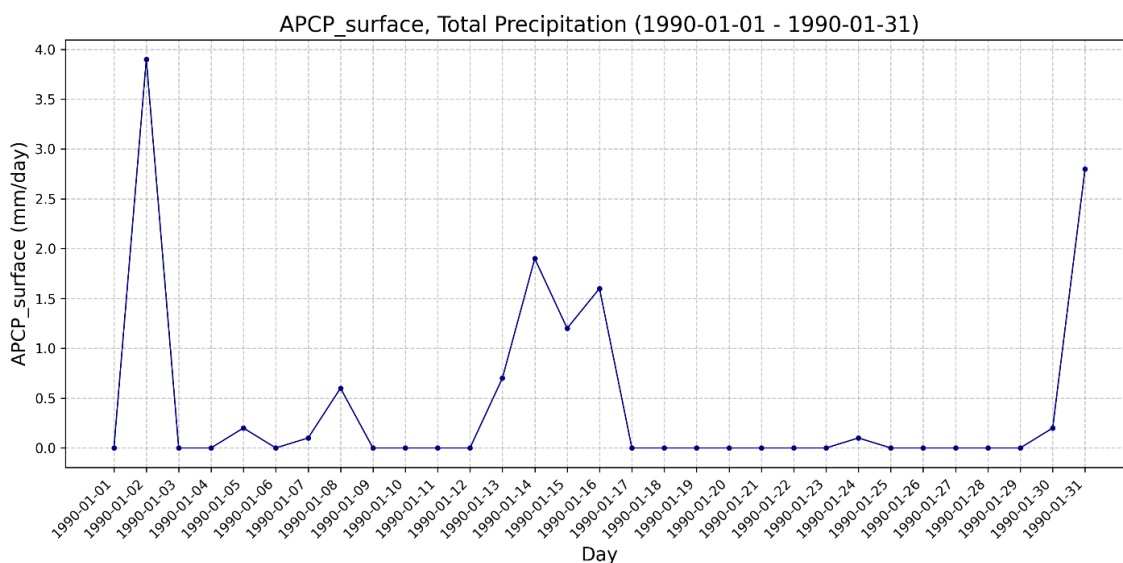


Figure 6. Example output from *AORC_LL_PointRetrieval.ipynb*: Total precipitation at a point near Salt Lake City, UT from January 1, 1990, to January 31, 1990, retrieved from the latitude-longitude gridded AORC dataset and aggregated into daily time intervals.

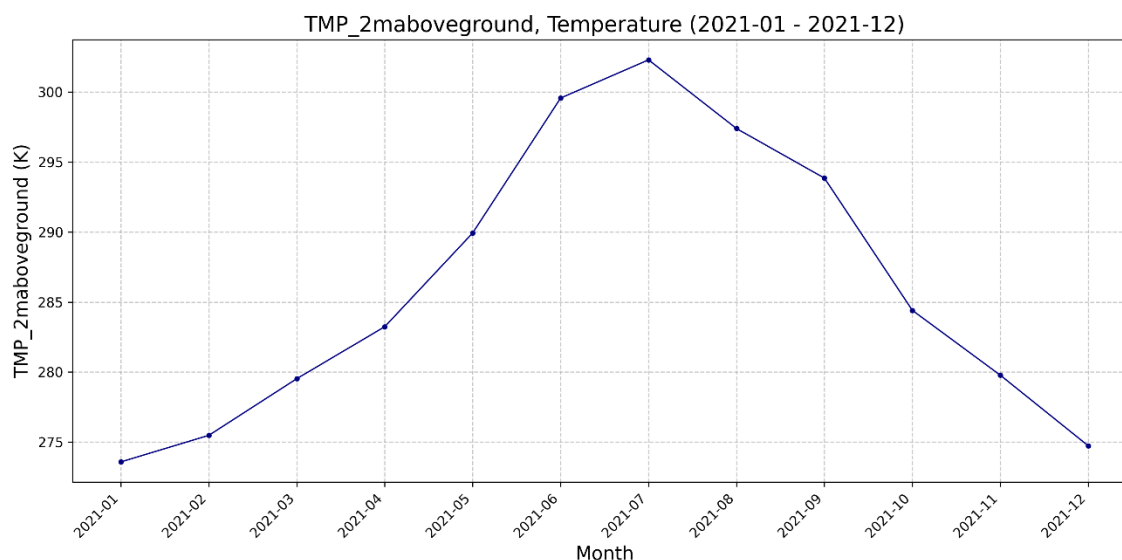


Figure 7. Example output from *AORC_LL_PointRetrieval.ipynb*: Temperature at a point near Salt Lake City, UT from January to December 2021, retrieved from the latitude-longitude gridded AORC dataset and aggregated into monthly time intervals.

3.2 Zone Data Retrieval Notebooks

AORC_LL_ZoneRetrieval.ipynb and *AORC_NWMPProj_ZoneRetrieval.ipynb*

The zone data retrieval notebooks were created to access AORC data for an area within the CONUS AORC coverage area, defined by a polygon shapefile. Each notebook takes as inputs the shapefile of a specific area, start and end dates for the desired period, a variable name, and a preferred time aggregation interval. The notebooks then retrieve data from Amazon Web Services (s3://noaa-nws-aorc-v1-1-1km for the latitude-longitude gridded dataset and s3://noaa-nwm-retrospective-3-0-pds/ for the NWM projected dataset), aggregate it over the specified time interval and individual shapes within the input shapefile, save the data as a comma separated values (CSV) file, and visualize it.

Figure 8 and Figure 9 illustrate example outputs from the *AORC_LL_ZoneRetrieval.ipynb* notebook. Figure 8 shows total precipitation across the Great Salt Lake Basin in January 2020, and Figure 9 indicates average total precipitation in the Great Salt Lake basin and subbasins from January 2020 to December 2021, retrieved from the latitude-longitude gridded AORC data and aggregated into monthly time intervals. The *AORC_NWMPProj_ZoneRetrieval.ipynb* notebook produces similar outputs, but the data are retrieved from the NWM projected dataset.

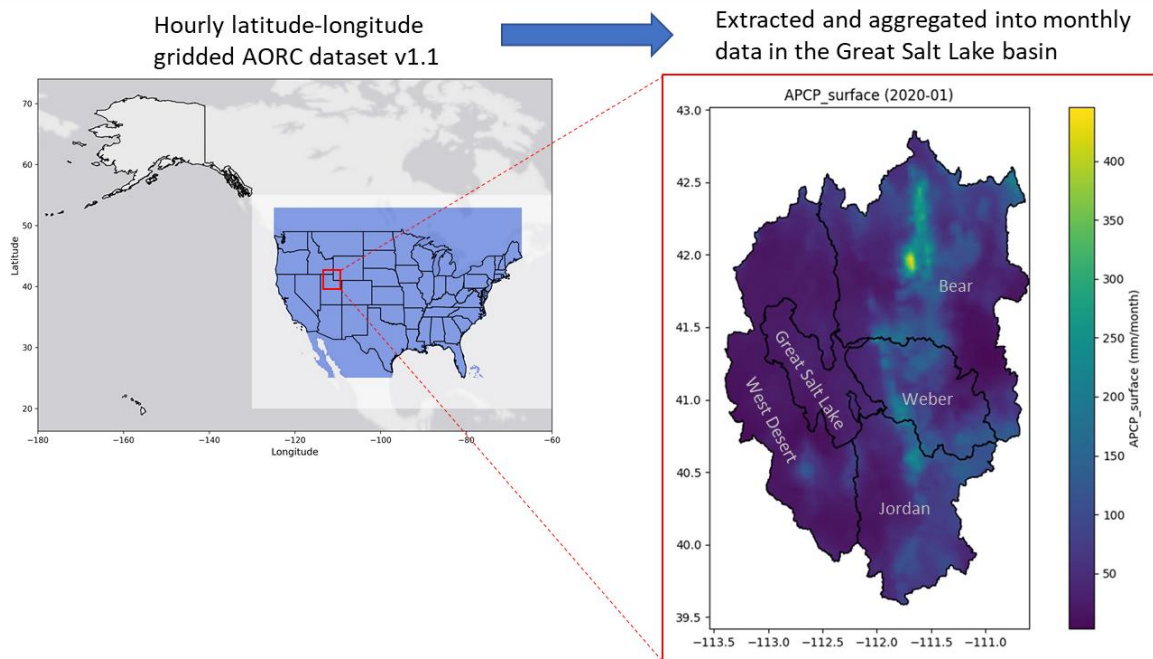


Figure 8. Example output from `AORC_LL_ZoneRetrieval.ipynb`: Total precipitation across the Great Salt Lake Basin in January 2020, retrieved from the latitude-longitude gridded AORC dataset and aggregated into a monthly time interval.

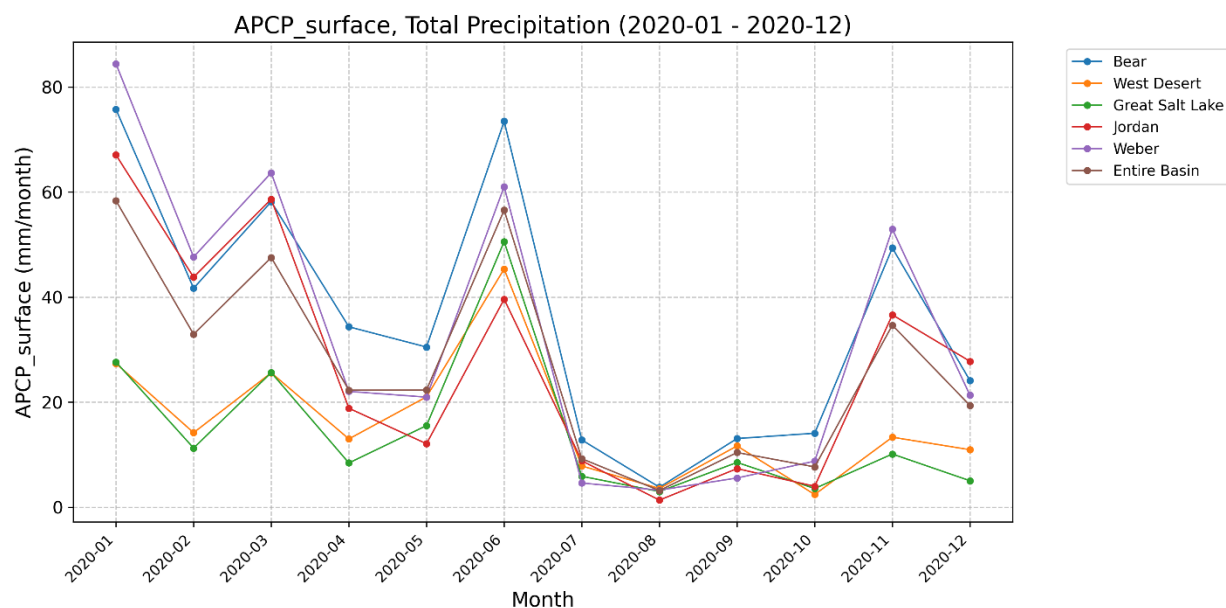


Figure 9. Example output from `AORC_LL_ZoneRetrieval.ipynb`: Average total precipitation in the Great Salt Lake basin and subbasins from January to December 2020, retrieved from the latitude-longitude gridded AORC dataset and aggregated into monthly time intervals.

4 Dataset Differences

There are small differences between the LL and NWM datasets. Some of these are illustrated here for a point near Salt Lake City airport, and for the Great Salt Lake basin. We compared the locations of the cell centers of these two datasets in ArcGIS (Figure 10) and analyzed the data retrieved for a point from each dataset for total precipitation (Figure 11 and Figure 12),

NWM projected data

- CRS: Projected, Lambert_Conformal_Conic (customized)

LL gridded data (NWS-AORC)

- CRS: EPSG:4326 - World Geodetic System 1984 (WGS84) geographic latitude and longitude reference system

Point of interest

- latitude = 40.77069
- longitude = -111.96503
- CRS: EPSG:4269 - Geodetic coordinate system for North America (gcs_north_american_1983)

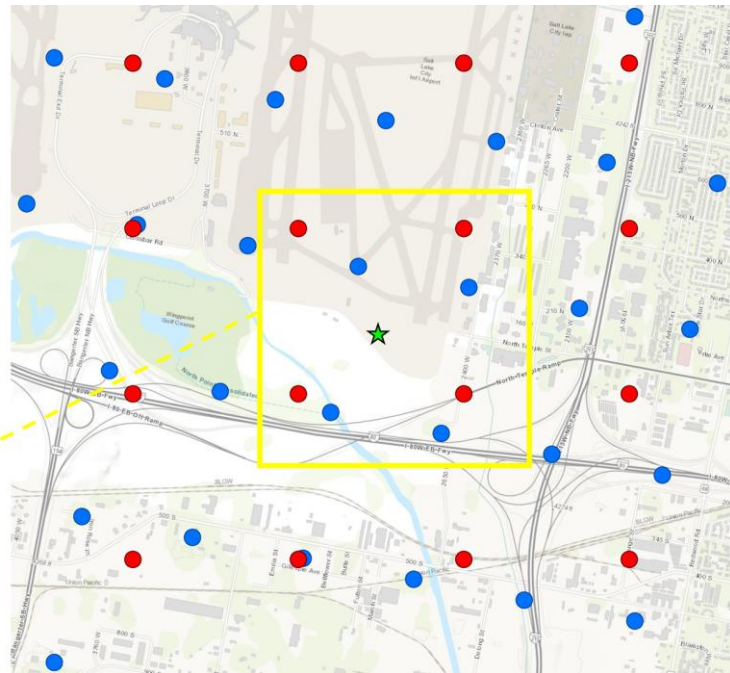
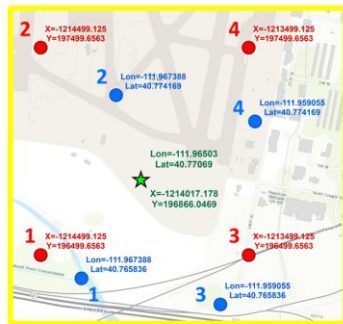


Figure 10. Map showing point of interest and the cell centers of the latitude-longitude gridded dataset (in blue) and the NWM projected dataset (in red). Points 1 to 4 represent the surrounding cell centers of each dataset around the point of interest.

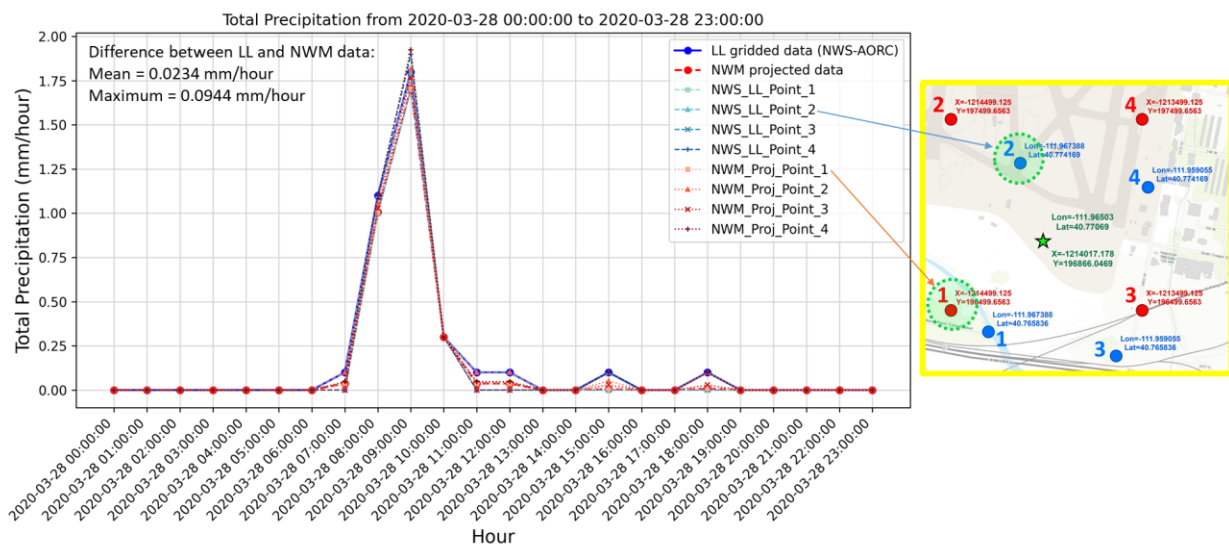


Figure 11. Hourly total precipitation of the points surrounding the point of interest, along with the data for the point of interest retrieved from the latitude-longitude gridded dataset (shown in light to dark blue) and the NWM projected dataset (shown in light to dark red).

Figure 11 shows the differences between the hourly precipitation data on March 28, 2020, retrieved for the point of interest from the latitude-longitude gridded dataset (in blue) and the NWM projected dataset (in red), as an example. The mean and maximum absolute differences are 0.02 mm/hour and 0.09 mm/hour, respectively. To investigate the causes of these differences and verify the accuracy of the data retrieval process, we also examined the data from the four surrounding points in each dataset (Figure 11). The analysis of the surrounding points indicates that, as the current code is set, the data from the nearest point to the point of interest in each dataset is retrieved as the data for the point of interest (point 2 for the LL dataset and point 1 for the NWM dataset, Figure 11). It is worth noting that for precipitation data, as we aggregate it over longer time intervals, such as years, the differences between the datasets accumulate, leading to larger discrepancies. To evaluate the impact of temporal aggregation, we retrieved total precipitation data and aggregated it to annual time interval (Figure 12). The mean and maximum absolute differences are 5.16 mm/year and 12.97 mm/year, respectively.

Figure 13 illustrates differences between the temperature data derived from the two datasets after 2016. Similar discrepancies are also observed in other variables (figures for the other variables are available in the appendix).

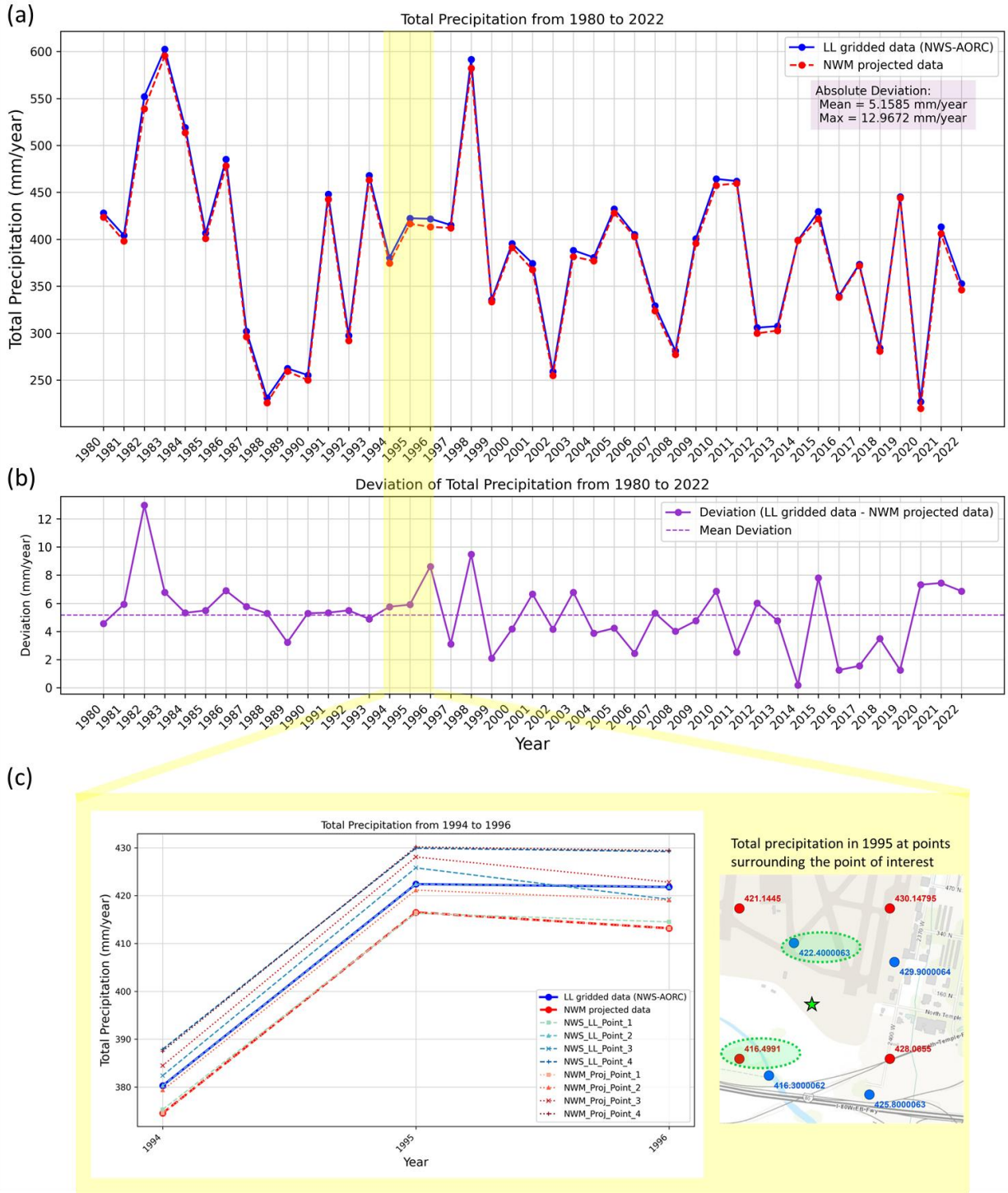


Figure 12. (a) Annual total precipitation retrieved from the latitude-longitude gridded dataset (in blue) and the NWM projected dataset (in red). (b) Differences between annual total precipitation data retrieved from the two datasets. (c) Zoom-in of the data for three years (1994 to 1996), showing the data for the surrounding points along with the nearest points to the point of interest.

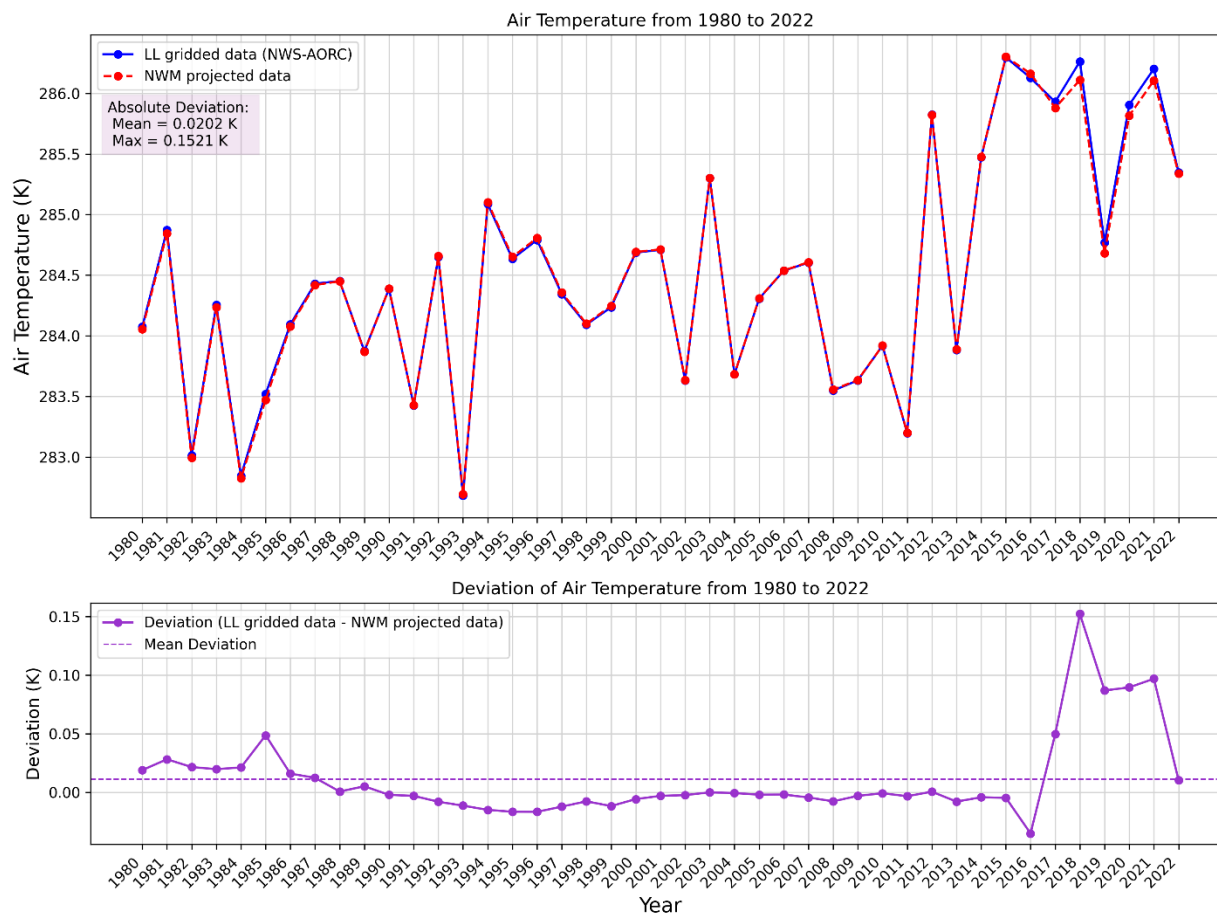


Figure 13. Annual air temperature retrieved from the latitude-longitude gridded dataset (in blue) and the NWM projected dataset (in red) and differences between them.

Figure 14 illustrates the precipitation in 1995 across the Great Salt Lake basin. The basin-wide average precipitation derived from the LL dataset (571.18 mm/year) and the NWM dataset (571.11 mm/year) are almost equal and show a very small and negligible difference. The averages over the subbasins are also very close, with the Jordan subbasin in the south showing the largest difference of approximately half a millimeter (0.42 mm).

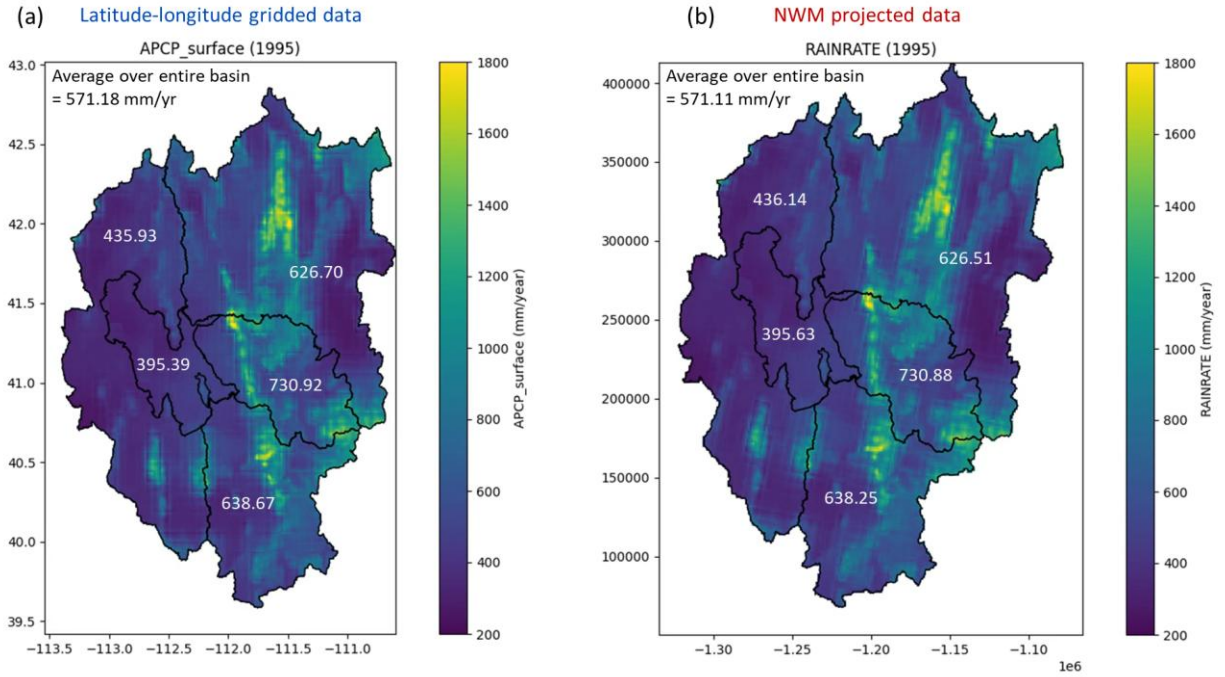


Figure 14. Total precipitation across the Great Salt Lake Basin in 1995. The data were retrieved from (a) the latitude-longitude gridded AORC dataset and (b) the NWM projected dataset, and aggregated both temporally and spatially to annual time intervals over the subbasins and the entire basin.

5 Summary and Conclusion

The AORC dataset is a valuable resource for hydrological and climate-related research within the CUAHSI and CIROH communities. The availability of two data sources (latitude-longitude gridded and NWM projected datasets) provides flexibility depending on the specific application. We developed Jupyter Notebooks to enhance the accessibility and processing capabilities for this data, enabling researchers to easily retrieve, subset, aggregate, and visualize AORC data for specific points or areas of interest. We compared these two datasets and found small differences between them, primarily related to projections. However, users should be aware of the potential discrepancies between the two datasets.

6 Appendix

Figures A1 to A6 compare the data for the other available variables retrieved from the two AORC datasets for the example point of interest near Salt Lake City, UT (see Figure 5).

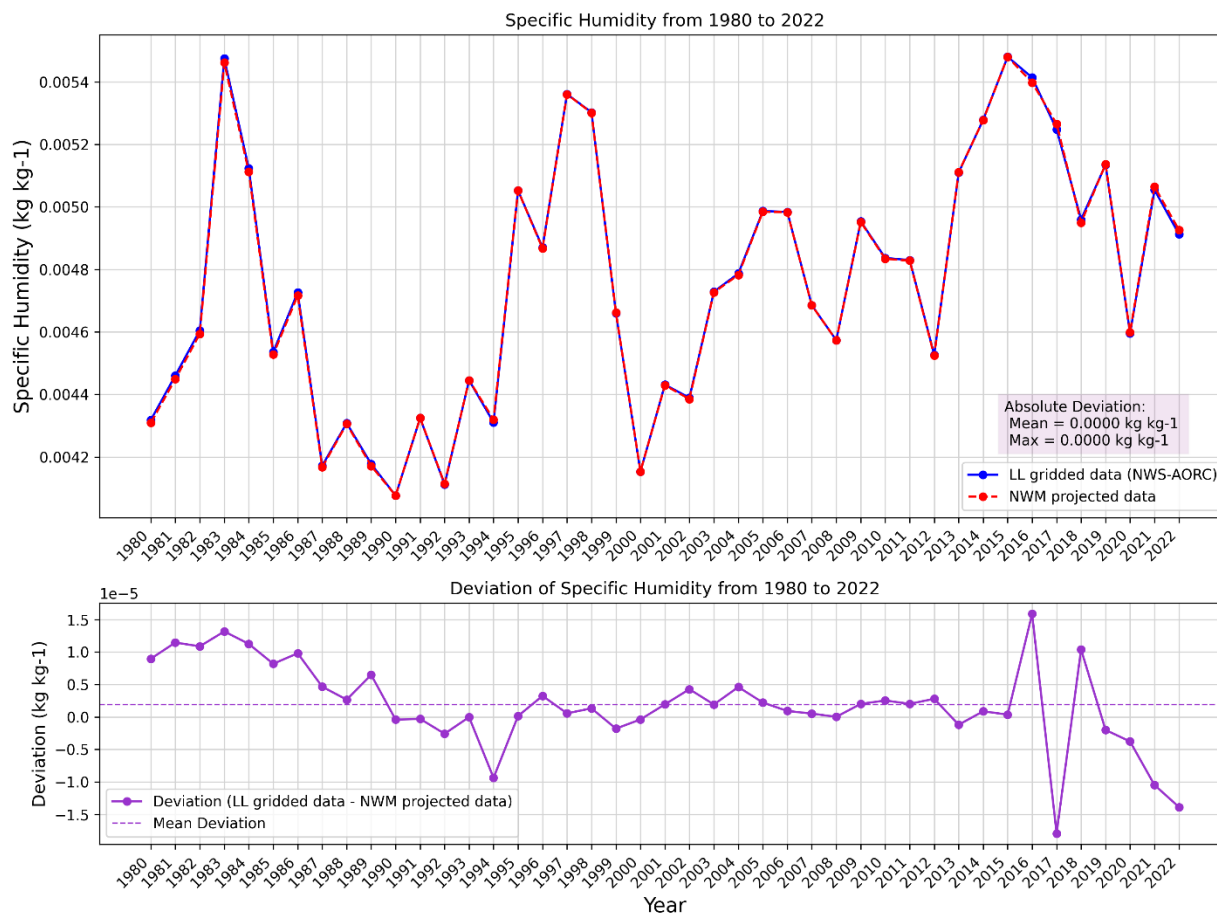


Figure A1. Specific Humidity data for the example point of interest near Salt Lake City, UT, retrieved from the latitude-longitude gridded dataset (blue line in the top plot) and the NWM projected dataset (red line in the top plot), along with the differences between them shown as the purple line in the bottom plot.

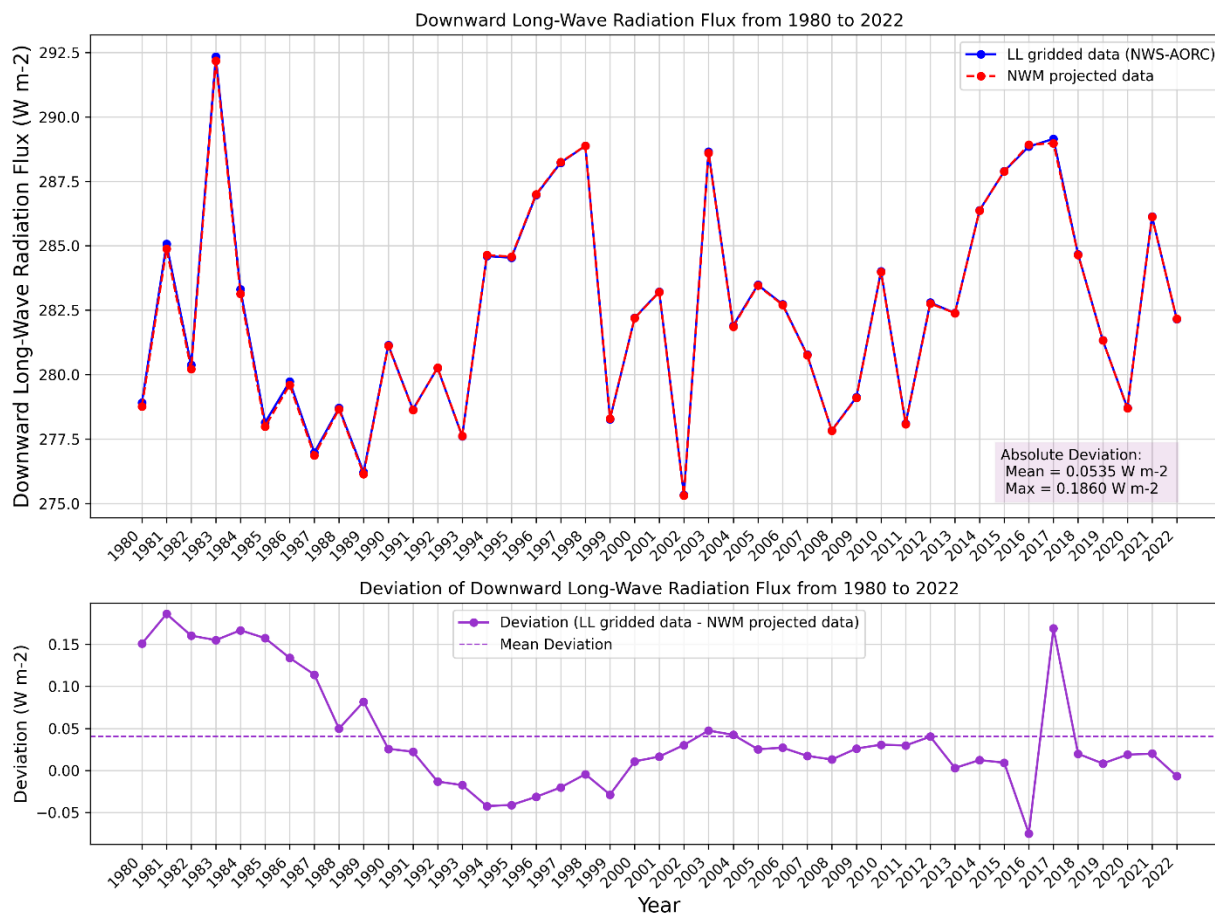


Figure A2. Downward Long-Wave Radiation Flux data for the example point of interest near Salt Lake City, UT, retrieved from the latitude-longitude gridded dataset (blue line in the top plot) and the NWM projected dataset (red line in the top plot), along with the differences between them shown as the purple line in the bottom plot.

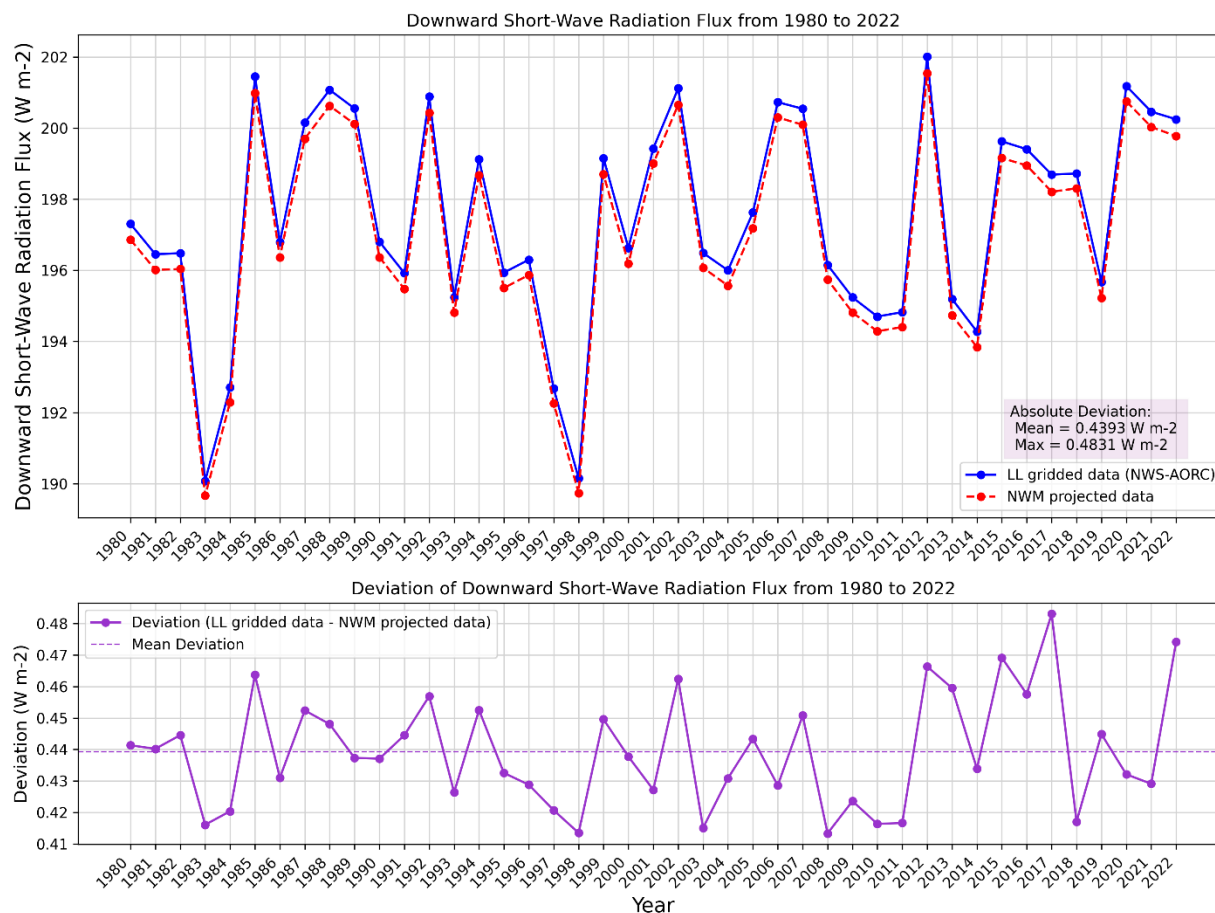


Figure A3. Downward Short-Wave Radiation Flux data for the example point of interest near Salt Lake City, UT, retrieved from the latitude-longitude gridded dataset (blue line in the top plot) and the NWM projected dataset (red line in the top plot), along with the differences between them shown as the purple line in the bottom plot.

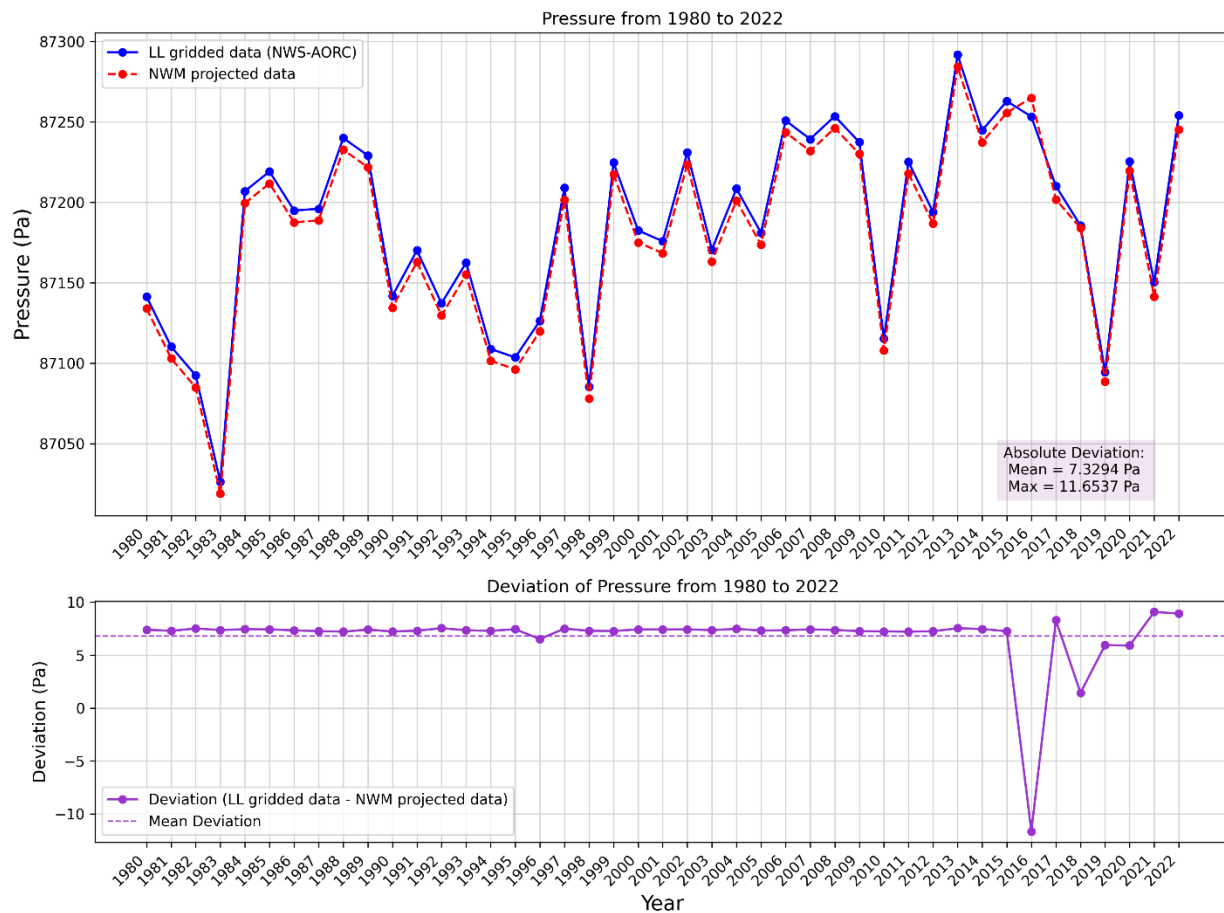


Figure A4. Air Pressure data for the example point of interest near Salt Lake City, UT, retrieved from the latitude-longitude gridded dataset (blue line in the top plot) and the NWM projected dataset (red line in the top plot), along with the differences between them shown as the purple line in the bottom plot.

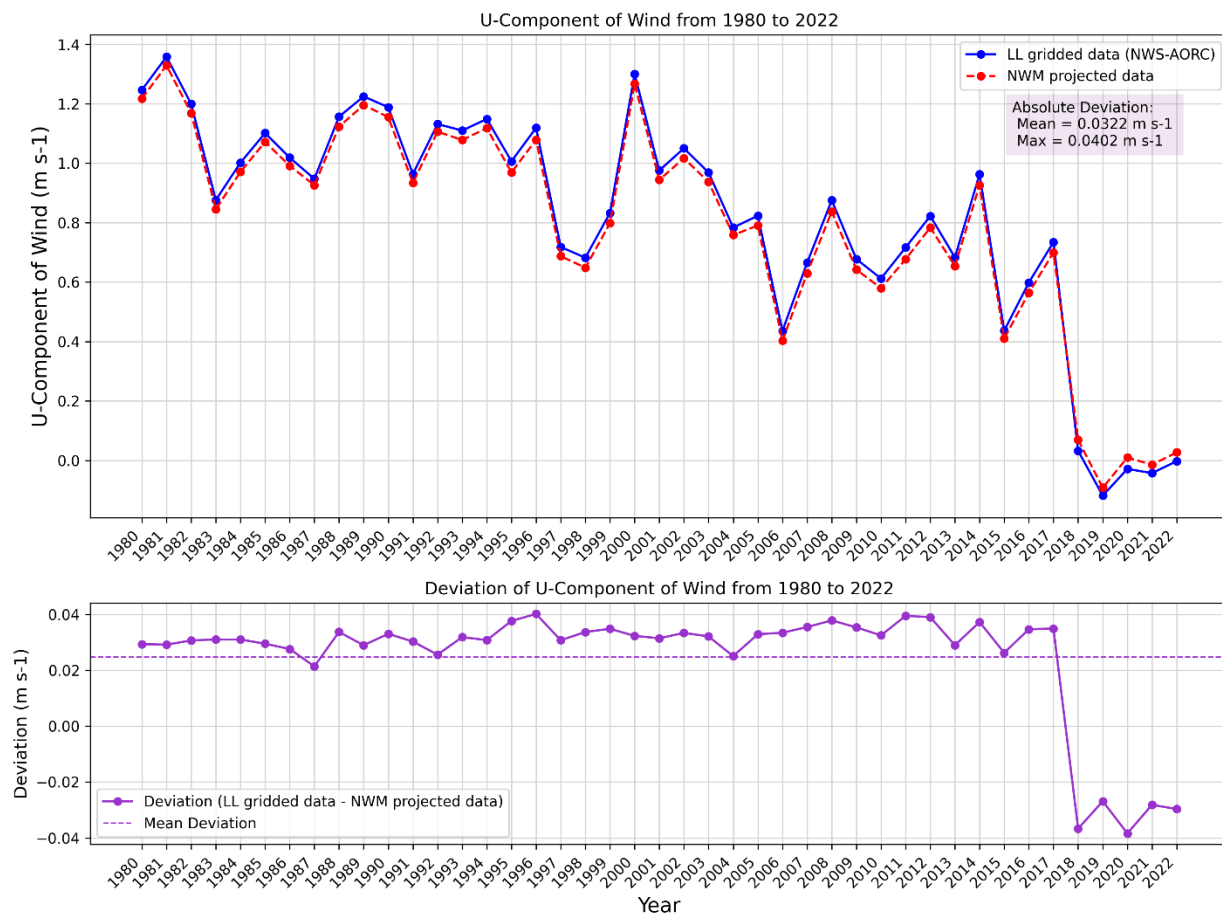


Figure A5. U-Component of Wind data for the example point of interest near Salt Lake City, UT, retrieved from the latitude-longitude gridded dataset (blue line in the top plot) and the NWM projected dataset (red line in the top plot), along with the differences between them shown as the purple line in the bottom plot.

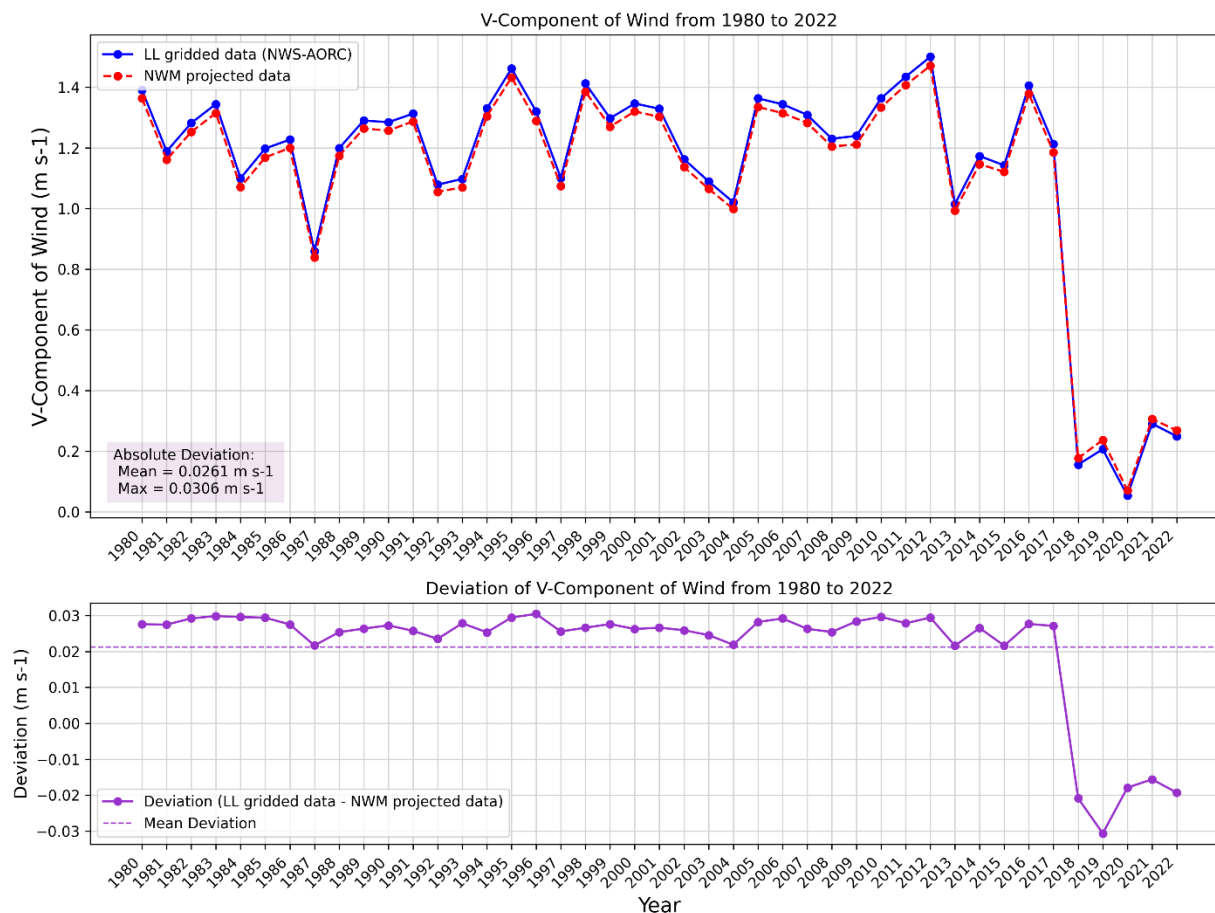


Figure A6. V-Component of Wind data for the example point of interest near Salt Lake City, UT, retrieved from the latitude-longitude gridded dataset (blue line in the top plot) and the NWM projected dataset (red line in the top plot), along with the differences between them shown as the purple line in the bottom plot.